1.1 Atomic Structure

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.1 Atomic Structure
Difficulty	Hard

Time allowed: 60

Score: /50

Percentage: /100

Question 1a

Fig. 1.1 shows the elements from the first three periods of the Periodic Table.

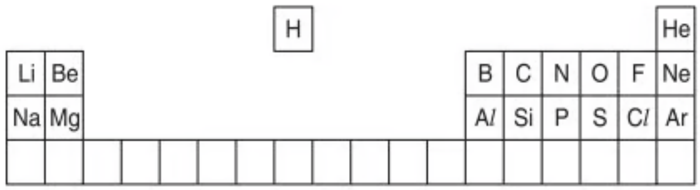


Fig. 1.1

Identify an element that fits each of the following descriptions:

i)

An element that forms a 2- ion with the same electronic configuration as Ne

[1]

ii)

The Period 3 element with the highest boiling point

[1]

iii)

The element from the first three periods with the largest atomic radius

[1]

iv)

The element from the first three periods with the highest first ionisation energy

[1]

v)

The Period 3 element with the successive ionisation energies shown in Table 1.1

Table 1.1

Ionisation number	1st	2nd	3rd	4th
Ionisation energy / kJ mol^{-1}	738	1451	7733	10541

[1]

[5 marks]

Question 1b

Fig. 1.2 shows the first ionisation energies for six consecutive elements labelled **A-F.**

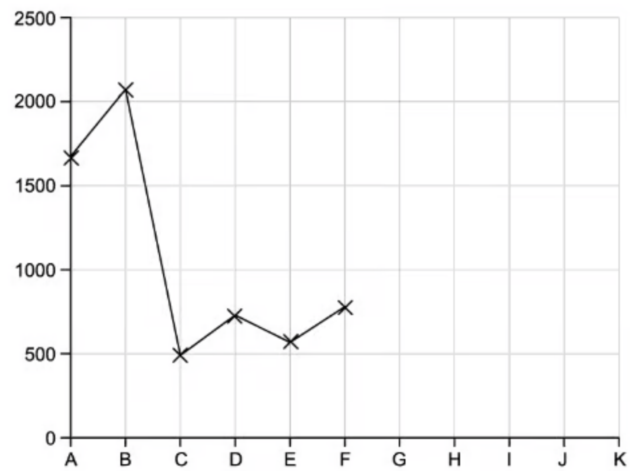


Fig. 1.2

Complete the graph to show the first ionisation energies of elements **G-K.**

[5 marks]

Question 1c

Explain why the value of the first ionisation energy for **D** is greater than for **C.**

[2 marks]

Question 2a

Successive ionisation energies provide evidence for the arrangement of electrons in atoms. In Table 2.1, the successive ionisation energies of oxygen are given.

Table 2.1

Ionisation number	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Ionisation energy / kJ mol^{-1}	1314	3388	5301	7469	10989	13327	71337	84080

i)

Give the equation, including state symbols, for the **third** ionisation energy of oxygen.

[2]

ii)

Explain how this data shows evidence of two energy shells in oxygen.

[2]

[4 marks]

Question 2b

Give the full electron configuration of the following atoms and ions.

i)
Te

[1]

ii)
Zn^{2+}

[1]

iii)
Cu^{2+}

[1]

[3 marks]

Question 2c

Palladium is a transition metal that is primarily used in a catalytic converter.

i)
Give the electron configuration for the Zirconium 2+ ion, Zr^{2+}, starting with [Kr].

[1]

ii)
Give the equation including state symbols to represent the third ionisation energy of Zirconium, Zr.

[1]

[2 marks]

Question 3a

Copper is an easily moulded base metal that is often added to precious metals to improve their elasticity, flexibility, hardness, colour, and resistance to corrosion.

Table 3.1 shows the number of protons, neutrons and electrons of different isotopes and ions of copper.

Table 3.1

Species	Protons	Neutrons	Electrons
		34	27
$^{65}Cu^{+}$	29		

i)

Complete Table 3.1.

[2]

ii)

Give the full electron configuration of the Cu^{+} ion.

[1]

[3 marks]

Question 3b

Calculate the percentage abundance of ^{63}Cu with a mass of 62.9296 and ^{65}Cu with a mass of 64.9278, when the average mass of the Cu isotope is 63.546.

Express your answer to an appropriate number of significant figures.

[3 marks]

Question 3c

Explain why the isotopes of copper exhibit the same chemical reactions but their densities differ.

[2 marks]

Question 3d

Copper can form a salt with chlorine to form copper(II) chloride. This reacts with sodium phosphate to form two different salts only.

Write a balanced equation for this reaction.

[2 marks]

Question 4a

Table 4.1 below shows the atomic radii for the elements of Period 2, Li to F.

Table 4.1

Element	Li	Be	B	C	N	O	F
Atomic radius / pm	152	112	88	77	70	66	68

i)

Explain the variation in atomic radius.

[4]

ii)

The value for neon is missing in Table 4.1. Explain why the atomic radius of neon cannot be measured in the same way as the other Period 2 elements.

[2]

[6 marks]

Question 4b

Complete Fig. 4.1 to show the electronic configuration of boron in the excited state.

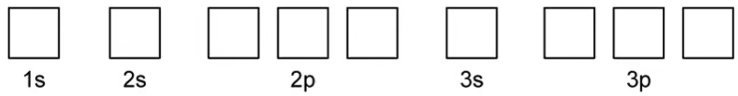


Fig. 4.1

[1 mark]

Question 4c

Explain why the first ionisation energy of boron is lower than the first ionisation of beryllium.

[2 marks]

Question 4d

The successive ionisation energies for another element, **J**, are shown in Table 4.2.

Table 4.2

Energy number	1st	2nd	3rd	4th	5th
Ionisation energy value / kJ mol^{-1}	738	1450	7733	10543	13630

State the formula of the compound when element **J** reacts with chlorine.

[1 mark]

Question 5a

In the blast furnace, carbon can react with oxygen to form carbon dioxide and carbon monoxide.

Write one equation for the this reaction.

[2 marks]

Question 5b

Identify and draw the subshell on Fig. 5.1 which has the highest occupied energy level in an oxygen atom.

Subshell …………………………….

[1]

Diagram

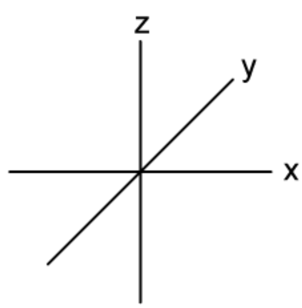


Fig. 5.2

[1]

[2 marks]

Question 5c

i)

Complete Fig. 5.2 to show the excited state of a carbon atom.

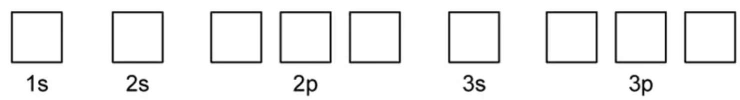


Fig. 5.2

[1]

ii)

Identify the type of hybridisation that arises in a molecule of carbon monoxide for both atoms and explain how this hybridisation occurs

Hybridisation ………….

Explanation

[4]

[5 marks]

Model Answers: Hard

1a

a) The elements are:

i) Oxygen/ O; [1 mark]

ii) Silicon/ Si; [1 mark]

iii) Sodium/ Na; [1 mark]

iv) Helium/ He; [1 mark]

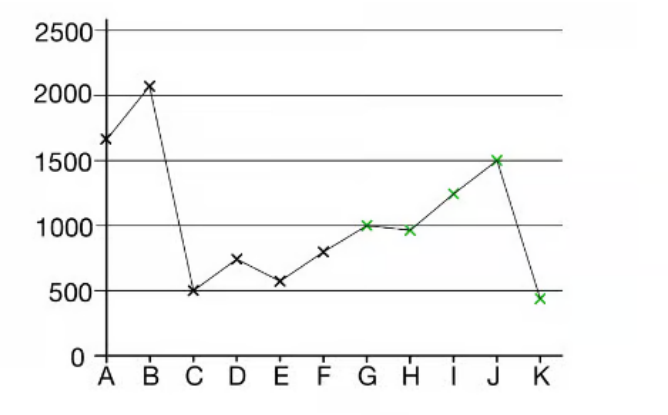
v) Magnesium/ Mg; [1 mark]

[Total: 5 marks]

- i) Oxygen has the electron configuration $1s^2\ 2s^2\ 2p^4$
  - Oxygen gains 2 electrons to form the oxide ion with a charge of 2-
  - The electronic structure of the oxide / O^{2-} ion is $1s^2\ 2s^2\ 2p^6$, which is the same as neon
- ii) Silicon has a giant covalent structure with lots of energy needed to overcome compared to:
  - Na - Al which have metallic bonding
  - P - Ar which have induced dipole-dipole / weak van der Waals forces
- iii) Sodium has a larger atomic radius than:
  - Elements in Periods 1 and 2 due to an extra energy level
  - The rest of the Period 3 elements due to sodium having a lower nuclear charge
- iv) Helium will have the highest first ionisation energy from H → Ar as:
  - It only has one shell so the outer electrons are closest to the nucleus
  - It has one extra proton compared to hydrogen
- v) You need to identify where the largest jump in ionisation energies is
  - In this case, the jump is between the second and third ionisation energies
  - So, the element is in Group 2 as a larger amount of energy is needed to remove an electron from the next shell down

1b

b) The completed graph is:



- G above F
  AND
  H between G and F; [1 mark]
- I above H and below A
  AND
  J above I and below B; [1 mark]
- K below C; [1 mark]

[Total: 3 marks]

- Element **B** must be the noble gas
  - This is because there is a large decrease after element **B**
  - This large decrease is consistent with the outer electron of **C** being in a shell further from the nucleus, which occurs at the start of a new period
- This means that element **C** is in Group 1
- There must be a drop in ionisation energy from **D** to **E** due to increased shielding (s to p shell)
- There must be another drop from **G** to **H** due to spin pair-repulsion

1c

c) The value of the first ionisation energy for **D** is greater than for **C** because:

- **D** has one more proton / has a higher nuclear charge

OR
The outer electron in **D** is closer to the nucleus; [1 mark]
- The electrons being removed from **D** and **C** are from the same (s) subshell; [1 mark]

[Total: 2 marks]

- This graph shows you that **C** is an element in Group 1 and **D** is an element in Group 2
- Element **B** must be the noble gas, as before there is a large decrease in energy value showing the outer electron is in a shell further away from the nucleus with the start of a new period

2a

a)

i) The equation for the **third** ionisation energy of oxygen is:

O^{2+} (g) → O^{3+} (g) + e^{-}

- Correct chemical formulae; [1 mark]
- Correct state symbols; [1 mark]

ii) Explanation of how this data shows evidence of two energy shells in oxygen:

- There is a significant / very large difference between 6th and 7th ionisation energies
OR
There is a significant / very large jump after the 6th ionisation energy; [1 mark]
- Which identifies a new shell / orbit / energy level that is closer to the nucleus; [1 mark]

[Total: 4 marks]

- **Remember:** Each successive ionisation energy after the 1st removes 1 mole of electrons from 1 mole of gaseous ions
- Large increases in successive ionisation energies identify the loss of a shell / orbit / energy level
    - The energy required to remove 1 mole of electrons from 1 mole of gaseous ions becomes far greater
    - This is due to the outermost electron now being much closer to the nucleus

2b

b) Give the full electron configuration of the following atoms and ions

i) Te

- $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^4$; [1 mark]

ii) Zn^{2+}

- $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$; [1 mark]

iii) Cu^{2+}

- $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9$; [1 mark]

[Total: 3 marks]

- When writing full electron configurations for atoms and ions you must begin from $1s^2$
- **Remember:** 4s and 5s subshells fill up before the 3d and 4d subshells
  - So, in order to write the electron configuration of the Te atom, apply the same process of filling up the shells as with atoms in lower periods
- **Remember:** The 4s subshell fills up first and also empties first
  - Therefore when Zinc, Zn, becomes a 2+ ion both 4s electrons are removed
- Copper, Cu, like Cobalt, Co, is more stable with a half-full 3d subshell and full 4s subshell
  - The electronic configuration for copper is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$
  - So the 1+ ion will be $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$ and the 2+ ion will be $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9$

2c

c)

i) The electron configuration for the Zirconium 2+ ion, Zr^{2+} starting with [Kr]:

- [Kr] $4d^2$; [1 mark]

ii) The equation including state symbols to represent the third ionisation energy of Zirconium, Zr.

- Zr^{2+} (g) → Zr^{3+} (g) + e^{-}; [1 mark]

[Total: 2 marks]

- To work out the electron configuration of a transition metal that is not in the first series follow the same principles
  - The s subshell fills up first and empties first
- Therefore for Zirconium / Zr, there will be no electrons present in the 5s subshell as this subshell empties first
- In order to write the third ionisation energy, the equation must begin with an X^{2+} ion
  - **Remember:** Only 1 mole of electrons must be removed for each given ionisation energy

3a

a)

i) Table 3.1 should be completed as follows:

Species	Protons	Neutrons	Electrons
$^{63}Cu^{2+}$	**29**	34	27
$^{65}Cu^{+}$	29	**36**	**28**

- [1 mark] for each correct row

ii) The full electron configuration of the Cu^{+} ion is:

- $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $3d^{10}$; [1 mark]

[Total: 3 marks]

- When calculating the number of protons neutrons and electrons look at two points
  - Is the species an ion?
    - If so, allocate the number of electrons accordingly (add extra electrons for negative ions and subtract for positive ions)
  - Take note of the atomic number in the periodic table
    - This will tell you the number of protons (this doesn't change between isotopes)
- Electronic configuration questions are often paired with atomic structure questions

- Copper is a transition element and the full electron configuration for an atom of copper is:
  - $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^1$, $3d^{10}$
- The electron configuration of the copper, Cu, atom is unusual as it is more stable with a full 3d subshell and half-full 4s subshell, so does not start with $4s^2$
- Chromium, Cr, also behaves in this manner. The electron configuration is:
  - $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^1$, $3d^5$
- It is acceptable to write either the 4s subshell or 3d subshell first

3b

b) The % abundance of ^{63}Cu and ^{65}Cu is:

- $63.546 = (62.9296 \times x) + (64.9278 \times (1 - x))$; [1 mark]
- $1.9982 \times x = 1.3818$; [1 mark]
- $0.69152 = 69.152\ \% = {}^{63}Cu$

AND

- $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $3d^{10}$; [1 mark]

[Total: 3 marks]

- When calculating the number of protons neutrons and electrons look at two points
  - Is the species an ion?
    - If so, allocate the number of electrons accordingly (add extra electrons for negative ions and subtract for positive ions)
  - Take note of the atomic number in the periodic table
    - This will tell you the number of protons (this doesn't change between isotopes)
- Electronic configuration questions are often paired with atomic structure questions
- Copper is a transition element and the full electron configuration for an atom of copper is:
  - $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^1$, $3d^{10}$
- The electron configuration of the copper, Cu, atom is unusual as it is more stable with a full 3d subshell and half-full 4s subshell, so does not start with $4s^2$
- Chromium, Cr, also behaves in this manner. The electron configuration is:
  - $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^1$, $3d^5$
- It is acceptable to write either the 4s subshell or 3d subshell first

3b

b) The % abundance of ^{63}Cu and ^{65}Cu is:

- 63.546 = (62.9296 $\times$ x) + (64.9278 $\times$ (1 - x)); [1 mark]
- 1.9982 $\times$ x = 1.3818; [1 mark]
- 0.69152 = 69.152 % = ^{63}Cu

  AND

  30.848 % = ^{65}Cu

  AND

  Both final answers given to 5 significant figures; [1 mark]

[Total: 3 marks]

- The overall abundance of the isotopes is 100 % so:
  - % of ^{63}Cu + % of ^{65}Cu = 100 %
- We can then change this to *abundance of* ^{63}Cu + *abundance of* ^{65}Cu = *1*

    - Rearranging this gives *abundance of ^{63}Cu = 1 - abundance of ^{65}Cu*
- Then you can continue with the mathematical equation and calculate the abundance of the ^{63}Cu isotope, then you can calculate the abundance of the ^{65}Cu isotope
    - $(0.69152 \times x) - (64.9278 \times x) = 63.546 - 64.9278$
    - $\frac{1.3818}{1.9982} = 0.69152$
- 0.69152 as a percentage is 69.152 % and is the abundance of the ^{63}Cu isotope
- 1 - 0.69152 = 0.30848, so 30.848 % is the percentage of the ^{65}Cu isotope
- You should always give the same significant figures as the value in the question with the least number of significant figures

3c

c) The chemical reactions are similar but densities are different because:

- There are the same number of electrons
  OR
  They have the same electronic structure; [1 mark]
- But, the mass number differs
  OR
  But, the number of neutrons is different; [1 mark]

[Total: 2 marks]

- Isotopes of the same element will have slightly different physical properties but the same chemical properties
    - Chemical properties are determined by the number of electrons
        - The number of electrons is the same in any isotope of an element
- Neutrons are neutral subatomic particles so they only add mass to the atom
    - As a result of this, isotopes have different physical properties such as small differences in their mass, density, melting point and boiling point

3d

d) The balanced equation for the reaction between copper(II) chloride and sodium phosphate is:

$$2Na_3PO_4 + 3CuCl_2 \rightarrow 6NaCl + Cu_3(PO_4)_2$$

- Correct reactant formulae
  AND
  Correct product formulae; [1 mark]
- Correct balancing; [1 mark]

[Total: 2 marks]

- This question does not give you many clues, but the key one is the fact the reaction produces two salts
  - So, there are only two products
- The trickier formula is copper(II) phosphate
  - The copper ion has a charge of 2+, the phosphate ion has a charge of 3-
  - Therefore, 3 copper ions and 2 phosphate ions are required to form a neutral compound
  - **Careful:** Don't forget your brackets
- In terms of balancing
  - Begin with the Na atoms
    - $\mathbf{2}Na_3PO_4 + CuCl_2 \rightarrow \mathbf{6}NaCl + Cu_3(PO_4)_2$
  - Then Cu atoms
    - $2Na_3PO_4 + \mathbf{3}CuCl_2 \rightarrow 6NaCl + Cu_3(PO_4)_2$
  - Each side of the equation now has
    - 6 Na, 3 Cu, 2 P, 8 O and 6 Cl

4a

a)

i) The variation is due to:

- The nuclear charge increasing (across the Period); [1 mark]
- There is similar shielding
  OR
  The electrons are in the same shell; [1 mark]
- There is a stronger attraction between protons and electrons; [1 mark]
- The outer electrons are / the outer shell is drawn in more by the increased nuclear charge; [1 mark]

ii) A value for neon cannot be measured in the same way because:

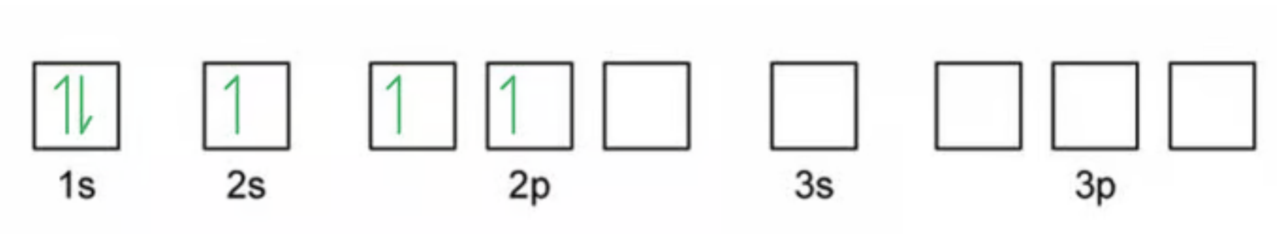
- Neon does not form bonds / compounds
  OR
  Neon exists as single atoms / is monoatomic; [1 mark]
- Atomic radius is half the distance between the nuclei of two covalently bonded atoms of the same type; [1 mark]

[Total: 6 marks]

- You need to make sure you can explain the difference in the following properties between elements:
  - Atomic radius
  - Ionisation energy
  - Melting/ boiling point
- The metallic or covalent radius is worked out by measuring the distance between the two nuclei (d) and dividing the result by 2
  - Atomic radius = $\frac{d}{2}$
- When measuring the atomic radius of a Group 0 atom, the atoms will only be touching each other as they do not form bonds
  - There are only van der Waals forces between these atoms
  - There are no covalent or metallic bonds

4b

b) The box notation of boron in the excited state is:

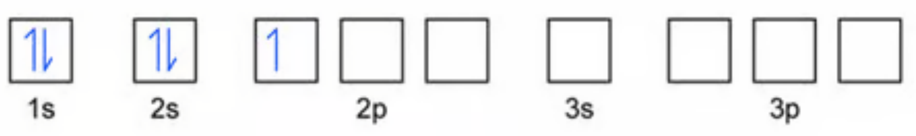
- 

  ; [1 mark]

[Total: 1 mark]

- The electronic configuration of boron is $1s^2\ 2s^2\ 2p^1$
  - Always write this out first before drawing the half-headed arrows in the box
- **Careful**: The question has asked for the electronic configuration of boron in the excited state

- Therefore, one 2s electron has been promoted to the 2p subshell
- The ground state electronic configuration would be:



4c

c) The first ionisation energy of boron is lower than that of beryllium because:

- The outer electron of Be is in a 2s subshell
  AND
  The outer electron of B is in a 2p subshell; [1 mark]
- A 2s subshell is closer to the nucleus than 2p
  AND
  Which means that more energy is required to remove the (Be) outer electron; [1 mark]

[Total: 2 marks]

- Boron has one electron in the 2p subshell, beryllium does not
- The electronic configurations for Be and B are:
  - Be: $1s^2$ $2s^2$
  - B: $1s^2$ $2s^2$ $2p^1$
- The 2p subshell electrons are higher in energy and are also shielded by the 2s subshell
  - Therefore, it is easier to remove them as they can more strongly resist the effective nuclear charge of the nucleus

4d

d) The formula is:

- JCl_2 ;[1 mark]

[Total: 1 mark]

- When trying to identify which group an element is in based on ionisation data, look for

the biggest jump
- E.g. If the largest jump is between the third and fourth ionisation energies, then the element is in Group 3
- In this case, the largest jump is between the second and third ionisation energies so element **J** is in Group 2

- If the element is in Group 2, the charge on the ion will be 2+
  - Two chloride ions would be needed to cancel out the charge of the metal ion, so the formula is JCl_2

5a

a) The equation for this reaction is:

$3C (s) + 2O_2 (g) \rightarrow CO_2 (g) + 2CO (g)$

- Correct reactants
  AND

4d

d) The formula is:

- JCl_2 ;[1 mark]

[Total: 1 mark]

- When trying to identify which group an element is in based on ionisation data, look for the biggest jump
  - E.g. If the largest jump is between the third and fourth ionisation energies, then the element is in Group 3
  - In this case, the largest jump is between the second and third ionisation energies so element **J** is in Group 2
- If the element is in Group 2, the charge on the ion will be 2+
  - Two chloride ions would be needed to cancel out the charge of the metal ion, so the formula is JCl_2

5a

a) The equation for this reaction is:

$3C (s) + 2O_2 (g) \rightarrow CO_2 (g) + 2CO (g)$

- Correct reactants
  AND
  Correct products; [1 mark]
- Correct balancing; [1 mark]

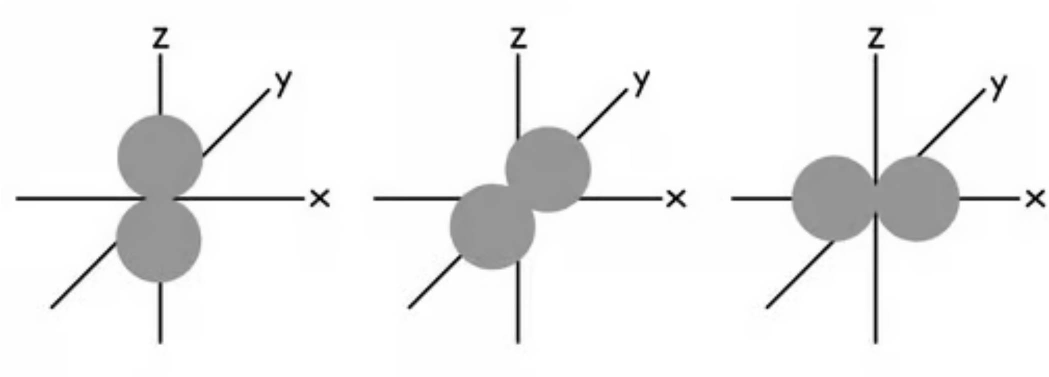
[Total: 2 marks]

- This reaction involves both complete and incomplete combustion as both carbon dioxide and carbon monoxide are formed
- You do not need to include state symbols in your equation, but it is good practice to do so

5b

b) The subshell which has the highest occupied energy level in an oxygen atom is:

- Subshell = 2p; [1 mark]



- Correct shape for p subshell; orbital can be drawn along any axis; [1 mark]

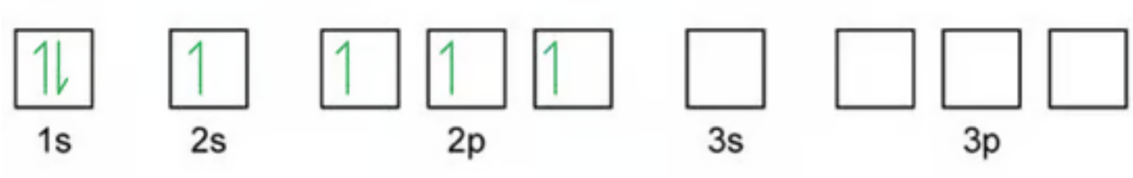
[Total: 2 marks]

- The electronic configuration of oxygen is
  - $1s^2 2s^2 2p^4$
  - So, the outermost electron is in the 2p subshell
- Make sure you say 2p subshell as this is the highest
- You should know the shapes of the s and p subshells

5c

c)

i) The excited state of a carbon atom is:

- 

; [1 mark]

ii) The type of hybridisation that arises is:

- Hybridisation = sp for both carbon and oxygen; [1 mark]

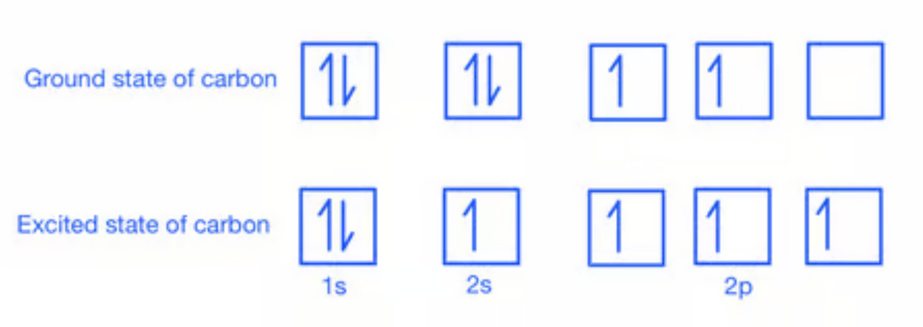
Explanation:

- CO has a triple bond; [1 mark]
- One sigma and two pi / π bonds formed; [1 mark]

- 2 unhybridised p orbitals and 2 sp hybridised orbitals formed; [1 mark]

[Total: 5 marks]

- The ground and excited state of carbon is:



- The carbon atom forms one σ bond and two π bonds with the oxygen atom
- One sp orbital forms a σ bond and one sp orbital occupies the lone pair
- The two unhybridised p orbitals form the π bonds